

A MODEL OF OPTIMAL SCIENCE FUNDING: TARGETING THE CAPABILITY-RESOURCE GAP

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ABSTRACT. The grant funder’s challenge is to maximize the impact of a fixed budget under uncertainty about the researchers who might receive it. We develop a Bayesian model of this challenge: researchers differ in their capabilities and in their resources; research output requires both, and funding compounds capability across rounds. We show that the optimal allocation funds the capability-resource gap: grants increase with capability and decrease with existing resources. Track records underdetermine the gap; peer review supplies an additional, noisy signal of capability. We find that this signal’s value increases with capability inequality and with the signal’s informativeness, and that under heavy-tailed capability distributions even a noisy signal from peer review captures most of that value. We further find that uniform seed grants and lotteries, which forgo targeting, lose little research output exactly where review is worth little, and that whom to fund matters far more than when to fund.

Significance. Research funders divide fixed budgets among researchers whose abilities and needs they cannot observe. Whether to concentrate funds among a few researchers or spread them across many, whether peer review is worth its cost, and whether lotteries should replace it are contested, and the evidence has not settled them. We model funding as a sequential decision under uncertainty and show that the optimal allocation funds the gap between each researcher’s capability and their resources, that track records alone underdetermine that gap, and that what peer review is worth, how lotteries and seed grants perform, and whether to concentrate or spread turn on two features of a field: the funder’s budget relative to researchers’ resources, and how unequal capability is.

The United States spent close to a trillion dollars on research and development in 2024, roughly a fifth of it funded by the federal government (National Center for Science and Engineering Statistics 2026; National Science Board 2026), and how best to allocate that money remains unsettled. Whether funding should be

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This work was supported by the John Templeton Foundation [grant number TODO]..

concentrated among a few researchers or spread across many is a long-running dispute (Aagaard et al. 2020). How well peer review predicts research outcomes is contested: among funded NIH grants, percentile scores predicted subsequent productivity barely better than chance (Fang et al. 2016), while studies with other designs find modest but real predictive power (Li and Agha 2015; Park et al. 2015). And several funders now allocate part of their budgets by lottery (Liu et al. 2020; Heyard et al. 2022).

These disputes concern one problem: what a funder with a fixed budget and imperfect knowledge of researchers should do. The empirical studies each measure one input to that decision, how well scores predict outcomes among funded grants or how research output responds to grant size, and none can say what the funder should do with what it knows; that depends on what the funder is uncertain about, what its information is worth, and how the answer changes with the budget. We treat the funder’s problem as a sequential decision under uncertainty and analyze it using a Bayesian approach. Researchers differ in capability, what they know and can do, and in resources, what they have to work with; output requires both and is limited by the scarcer of the two; the funder observes output, and the noisy judgments of reviewers, but neither attribute directly. A grant adds resources in the round it is given and, because doing research builds expertise, raises capability in later rounds. Each round, the funder observes the output that results, updates its beliefs about the researchers, and allocates again.

Three results follow. First, the optimal allocation funds the gap between each researcher’s capability and their resources: grants increase with capability and decrease with existing resources, and the budget sets how many researchers are funded at all. Two heuristics a funder might use, funding the track record and funding the under-resourced, each track one of the two quantities, and each can perform worse than simply dividing the budget uniformly across all researchers. Second, track records underdetermine the gap, and peer review can provide a further signal of capability. The value of that signal, how seed grants and lotteries perform relative to targeted funding, and whether funds should be concentrated among a few researchers or spread across many all depend on the same two features of a field: how much of researchers’ resources the funder’s budget can supply, and how unequal capability is. Where the budget is tight and capability heavy-tailed, targeting is worth a great deal, even noisy review captures most of that value, and lotteries give up much of it; where the budget is ample and capability evenly spread, little research output is at stake and spreading funds widely loses almost nothing. No funding mechanism is good or bad in itself. Third, when to

fund matters far less than whom to fund: the gain from choosing the schedule of spending is smaller than the gain from a review signal in every setting we examine.

Two assumptions bound these results: grants are used up in the round they are given, with whatever lasts from them entering as capability, and capability and resources are complements. The model is deliberately simple. Its contribution is to the theoretical understanding of the funder’s problem: it shows that the three disputes turn on two features of a field that a funder can estimate, and it says which way each dispute goes in each kind of field.

RESULTS

The optimal allocation funds the capability-resource gap. Consider first a funder that knows every researcher’s capability K_i and resources R_i and divides one round’s budget B so as to maximize expected output, which for researcher i is $2AK_iR_i/(K_i + R_i)$ with R_i the sum of baseline resources R_{i0} and the grant g_i (Materials and Methods). The marginal value of a dollar to researcher i is $2AK_i^2/(K_i + R_{i0} + g_i)^2$: it increases with their capability and decreases with the resources they already hold, grants included. An allocation is optimal exactly when no dollar can be moved to a researcher who would produce more with it, and solving that condition gives the gap rule (SI Text 3, Proposition 1):

$$g_i^* = \max(cK_i - R_{i0}, 0),$$

where $c > 0$ is the value at which the grants sum to the budget. Each researcher’s target is c times their capability; the rule fills each positive gap between target and resources and grants nothing to the rest. The condition $R_{i0} < cK_i$ draws a frontier through the population, separating the funded from the unfunded (Fig. 1). When the budget is small, c is small and few researchers are funded; a larger budget raises c , moves the frontier outward, and funds more researchers with larger grants, until at a large enough budget funding extends across nearly the whole population.

The gap rule explains why each heuristic fails. Funding the track record fails because output reflects resources as well as capability: a productive, well-resourced researcher has a small or negative gap, and an additional dollar adds little to their output. Funding the under-resourced fails for the opposite reason: scarcity is no evidence of capability, and dollars given to researchers of low capability add little output however scarce their resources. Each heuristic can produce less output than dividing the budget equally (SI Text 3, Examples 1 and 2).

How much targeting matters depends on the budget. Expected output increases with resources at a diminishing rate and never exceeds $2AK_i$, so when the budget is large enough to bring every researcher’s output near that bound, the optimal

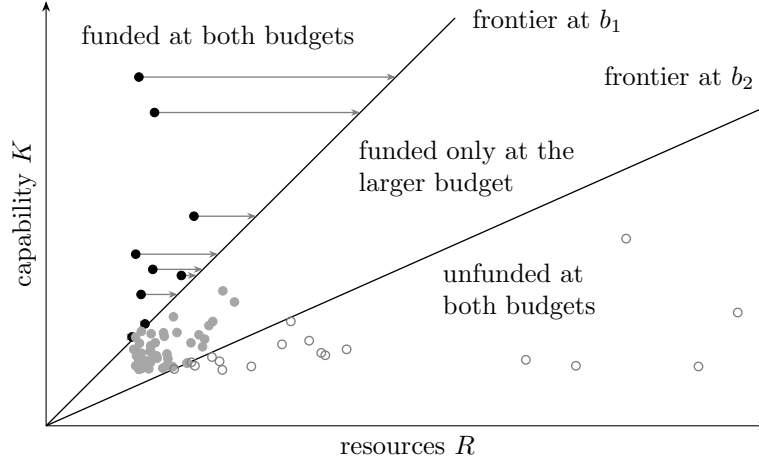


FIGURE 1. The gap rule in one population at two budgets. Each point is a researcher, placed by resources and capability. The two lines are the funding frontiers $R = cK$ at a smaller budget b_1 and a larger budget b_2 ; a larger budget raises c and flattens the frontier. Researchers to the left of a frontier are funded at that budget: black points at both budgets, gray points only at the larger budget, open points at neither. Each arrow is a grant at the smaller budget, which moves its researcher exactly to that budget's frontier; at the larger budget every black and gray researcher receives a grant that moves it to the outer frontier (arrows not drawn).

allocation and equal division differ little. The value of targeting, the expected output the optimal allocation adds over equal division, decreases in the budget and vanishes as the budget grows (SI Text 3, Corollary 2; Fig. S1). Targeting matters where the budget is small relative to researchers' resources, and this dependence on the budget recurs in every result below.

What the funder can learn: track records and peer review. A real funder observes research output, not capability or resources. Expected output is one number produced by the two jointly, so a researcher of high capability and modest resources and a researcher of modest capability and abundant resources can produce identical records, and these are exactly the researchers the gap rule treats most differently: the first may have the largest gap in the population, the second none. Output gives only a lower bound on capability, $K_i > \lambda_i/2A$. The problem is worst where the gap is largest. When resources are small relative to capability, expected output is approximately $2AR_i$, proportional to resources and nearly independent of capability (Fig. 2A), so on small grants output carries little information about capability, and only grants comparable to capability itself make records informative about it.

A Bayesian funder relying on records alone therefore allocates far from the gap rule, and further rounds of records improve its allocation only slowly: the correlation of its grants with the optimal grants rises from 0.08 in the first round to 0.41 by round twenty, still below the 0.81 that a funder with a review signal reaches in round one (Fig. 2*B*). The reason lies in the model’s structure. Capabilities compound while resources do not, so each researcher’s output approaches the level their resources sustain, and output at that level carries less and less information about capability.

Peer review can provide a further signal of capability: a noisy one, but one whose noise does not come from resources. We measure the value of review as the additional expected output of a funder allocating with the signal over the same funder allocating without it, and we report it as the share of the records-only shortfall that the signal recovers, where the records-only shortfall is the difference between the records-only funder’s output and the output a funder with complete information would produce. That value depends on the field (Fig. 2*C*). Review is worth most where capability is heavy-tailed and the budget is tight, and under heavy-tailed capability even a signal of moderate informativeness captures most of its value: at three times the default signal noise, review retains 82% of its maximum value when the capability distribution’s tail parameter is $\alpha_K = 1.3$, 44% at the default distribution ($\alpha_K = 2$), and 10% where capability is spread more evenly ($\alpha_K = 3.5$). The reason is twofold. Where capability is heavy-tailed, most of the output a funder can add comes from getting resources to the few researchers of unusually high capability, and precisely those researchers are the easiest to identify: they stand far apart from everyone else, so even a noisy signal separates them. Where capability is spread more evenly, no researcher matters much more than another, so review of any informativeness adds little; and where the budget is ample, information about whom to fund has little left to add. This ordering of review’s value by field holds across production functions from Cobb-Douglas to Leontief (SI Text 6). One condition qualifies it: the funder must weight review at its true informativeness. A funder that overtrusts review, treating a noisy signal as reliable, can do worse than one that ignores review altogether, and overtrusting review loses more output than undertrusting it (Fig. S2).

When to fund. The funder holds a fixed budget for the whole horizon and may spend it on any schedule. The strongest argument for spending early concerns fields whose researchers have few resources: a researcher without resources produces nothing, so the funder observes nothing and no capability grows, and money held for later rounds produces neither evidence nor growth in the meantime. The argument fails. A funder that spends everything in the first round allocates before observing any research output, and knows exactly as little as a funder that

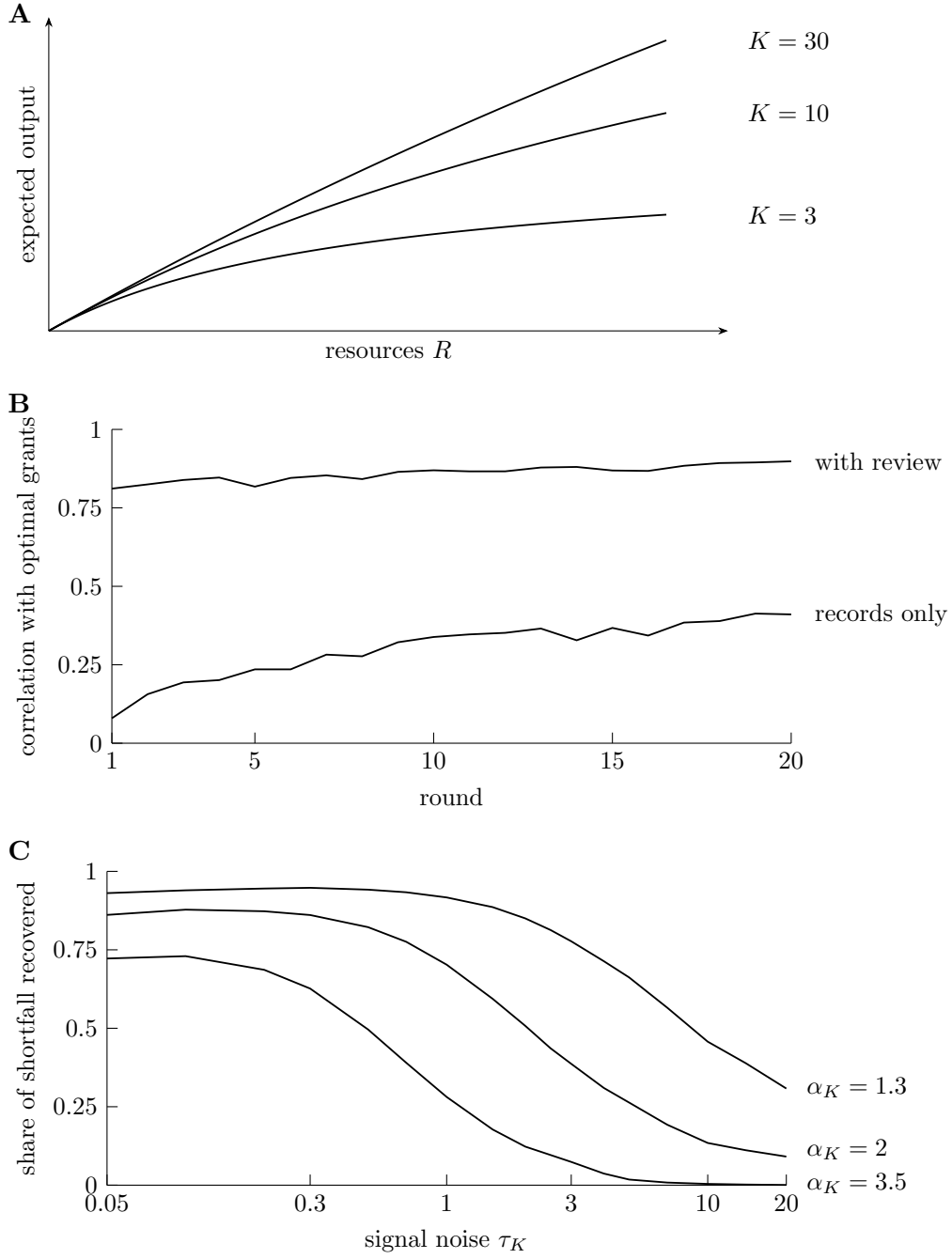


FIGURE 2. What the funder can learn. (A) On small grants, research output carries little information about capability: expected output against resources for researchers of capability $K = 3, 10$, and 30 ; the curves coincide where resources are small relative to capability. (B) Records do not catch up: the correlation of each funder's grants with the optimal grants, round by round; the records-only funder improves slowly and after twenty rounds remains further from the optimal allocation than the funder with the review signal is after one. (C) Review is worth most where capability is heavy-tailed: the share of the records-only shortfall that the review signal recovers, as review noise τ_K increases, for three capability distributions. Settings: SI Text 7.

spends everything in the last round; in a field with no resources at all, the two schedules produce identical expected output. What the argument does establish is a complementarity: early spending enables observation and growth, and late spending exploits what early spending revealed. That favors putting some money early. It does not favor putting most of it early.

Our simulations say which should come first: observation, then money. Wherever capabilities compound at more than a negligible rate (ϵ above about 0.03), the optimal schedule spends less than an even share in the first round and spends the bulk once output has revealed who is capable; scarce baseline resources raise the early share and weakens the effect, but does not reverse it (Fig. 3A). The shift of spending toward later rounds comes from the output researchers produce whether or not they are funded, which compounds and reveals capability while the funder waits (SI Text 5).

The choice of schedule is worth less than the signal provided by grant peer review, in every setting we test. Across all horizons and compounding rates in our sweeps, the gain from choosing the schedule, over spending equal installments each round, stays below the review signal’s value: by an order of magnitude or more at the default compounding rate and below, and approaching the signal’s value only at the highest compounding rate and the longest horizon we consider (Fig. 3B). As a share of the research output that the forward-looking funder’s funding adds, the scheduling gain ranges from indistinguishable from zero to 4.5% across the 30 settings tested; the review signal’s value in the same settings ranges from 7% to 27%. The gain exists only where capability and resources are complements: under Cobb-Douglas production it is indistinguishable from zero (SI Text 6). Choosing whom to fund outweighs choosing when.

Spreading funds: seed grants, lotteries, and concentration. Two alternatives to targeting recur in funding policy, and each has a rationale. Uniform seed grants give every researcher the same amount, so that a researcher whom review misjudges still receives something; in practice a funder gives out part of its budget as seed grants and targets the rest. Lotteries give grants to applicants chosen at random, usually from among those who pass a review threshold; their advocates argue that review cannot reliably rank the applicants it deems fundable, that review is costly, and that review is biased (Fang and Casadevall 2016; Roumbanis 2019). Both alternatives give up part of targeting on purpose, so both give up part of the value of targeting, and that value depends on the field. For a funder allocating with records and review, targeting adds more research output than uniform funding’s entire gain over no funding in a heavy-tailed field with reliable review and a tight budget, and about a quarter of that gain in the default

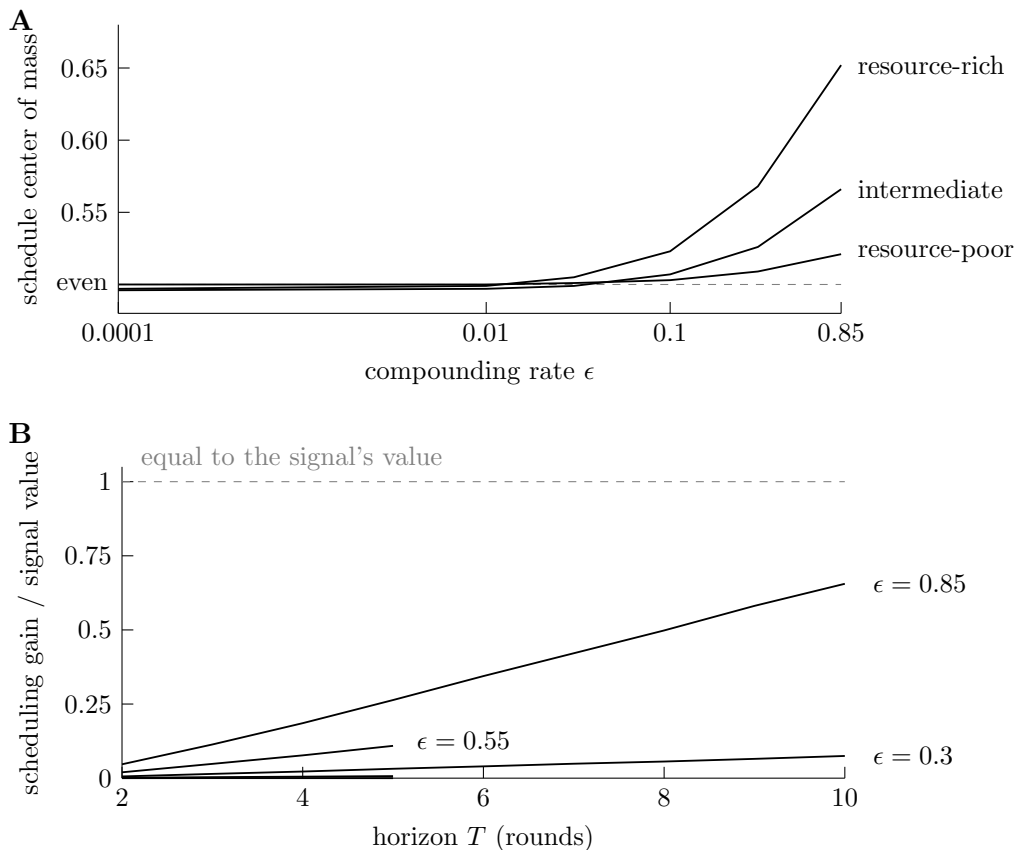


FIGURE 3. When to fund. (A) The compounding rate, not the scarcity of baseline resources, sets how far spending shifts toward later rounds: the schedule’s center of mass (0.5 is even; higher is later) as the compounding rate ϵ increases, for three baseline resource scales. (B) Choosing whom outweighs choosing when: the gain from choosing the schedule, as a share of the review signal’s value, by horizon and compounding rate; in every setting the gain is below the signal’s value. Settings: SI Text 7.

field at the largest budget we sweep (SI Text 5). What the alternatives save, in the cost of review and in the effort researchers spend on proposals, lies outside our model (Gross and Bergstrom 2019).

Consider seed grants first. The output lost increases faster than the fraction of the budget given out as seed grants: giving out half of the budget as seed grants loses more than twice the output that giving out a quarter loses (Fig. 4A). The size of the loss is set by what targeting is worth in the field: when a review-informed funder gives out three quarters of its budget as seed grants, it forfeits about 11% of the research output its funding adds in a heavy-tailed field with reliable review (17% at the smallest budget we sweep), and about 3.5% in the default field.

Now consider lotteries. A lottery gives each applicant in its pool a chance at a full grant instead of a certain smaller one, and on average the applicant receives the same amount either way. Because expected output increases with resources at a diminishing rate, the certain grant produces more expected output. Dividing a pool's money equally among the pool therefore produces more expected output than any lottery over the same pool, and a lottery over all applicants produces less than uniform funding (SI Text 4, Proposition 2). To measure how much less, and how lotteries compare with selection by review, we evaluate five schemes in a single round (Fig. 4B): equal division among the fifth of researchers with the highest review scores (selection by review); equal division among the top two fifths; the partial lottery that some funders run, which funds half of the top two fifths at random with full-size grants; uniform funding; and a lottery over all researchers with full-size grants. In a heavy-tailed field with a tight budget, selection by review gains about three quarters of what the optimal allocation gains when review is reliable and about half when review is very noisy. The partial lottery gains less, and most of the difference comes from ignoring the review ranking within the top two fifths rather than from the randomness itself: at the default noise, the partial lottery falls 0.18 short of selection by review, of which ignoring the ranking accounts for 0.14 and the randomness for 0.04. The lottery over all researchers gains least, less than uniform funding. In an evenly spread field with a large budget, the ordering reverses: uniform funding gains more than four fifths of what the optimal allocation gains, and every scheme that concentrates grants, whether by review score or by chance, gains less. A lottery among applicants above a review threshold therefore loses little output only where the review ranking it ignores is worth little: replacing that ranking by chance loses a few hundredths of the optimal allocation's gain in an evenly spread field or when review is very noisy, and about a sixth of that gain in a heavy-tailed field with reliable review. In an evenly spread field, the better alternative to review is small grants to many researchers; a lottery among a few applicants above a review threshold produces less.

Seed grants and lotteries both spread funds more widely than targeting does, and both raise the older question of whether funding should be concentrated among a few researchers or spread across many. The production function does not settle that question. Stronger complementarity between capability and resources concentrates the review-informed funder's grants when the budget is tight in a heavy-tailed field, spreads them when the budget is ample, and has a non-monotone effect where capability is evenly spread (Fig. S4; SI Text 5). How widely funds should be spread is set jointly by the production function, the budget, and the field's capability distribution.

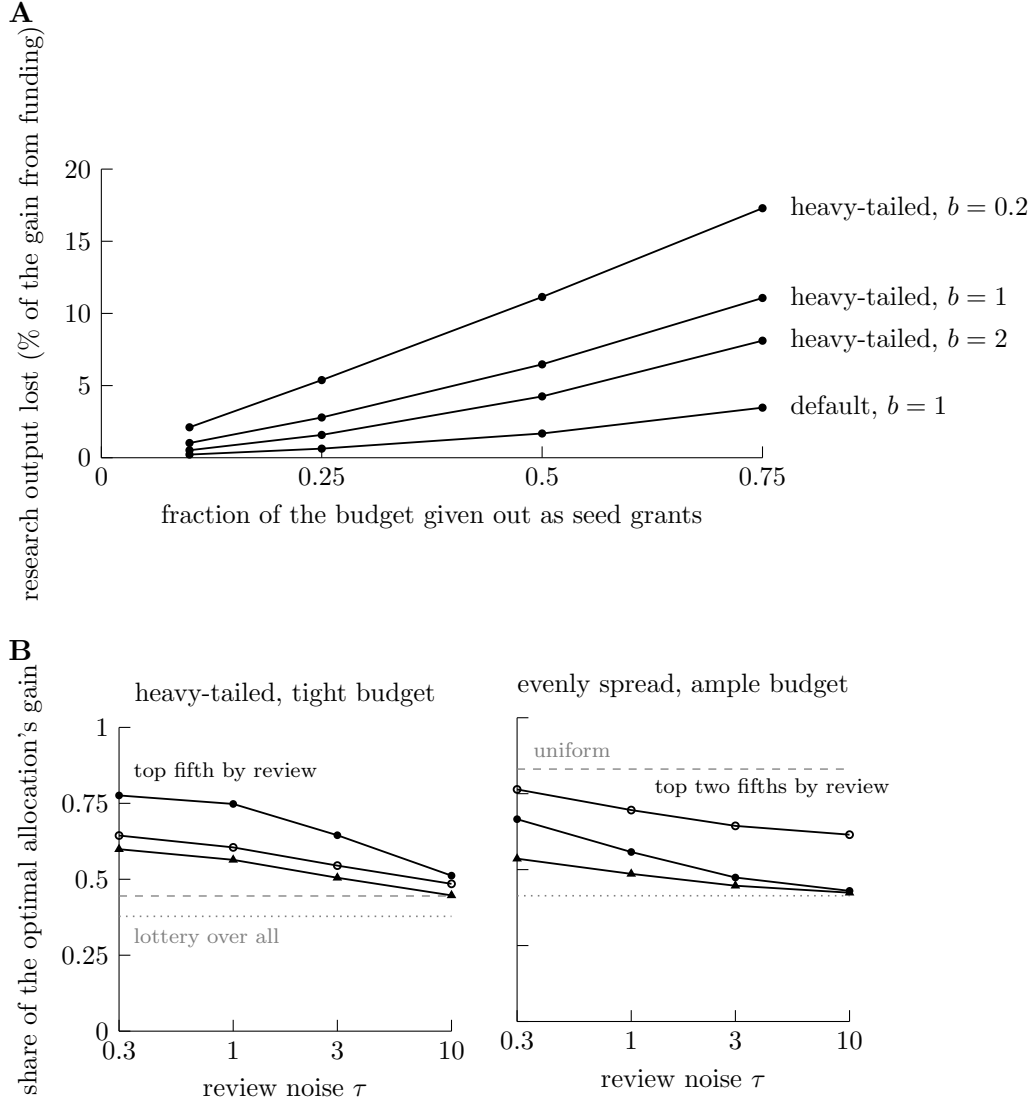


FIGURE 4. Spreading funds. (A) The research output lost to uniform seed grants, as a percent of the research output that the funder's grants add over no funding, when a review-informed funder gives out a fraction of its budget as equal grants to all researchers and targets the rest; the loss grows faster than the fraction, and its size is set by what targeting is worth in the field. (B) What five funding schemes gain, by field: each scheme's gain over no funding as a share of the optimal allocation's gain, as review noise increases, in a heavy-tailed field with a tight budget (left) and an evenly spread field with a large budget (right). Filled circles: equal division among the top fifth by review score; open circles: among the top two fifths; triangles: the partial lottery; dashed: uniform funding; dotted: a lottery over all researchers. Settings: SI Text 7.

DISCUSSION

Our results isolate one quantity as key to optimal grant funding: the value of targeting researchers with the greatest gap between their capabilities and their resources. Two features of a field determine this value. The first is the funder's budget relative to researchers' own resources: where the budget is small, dollars are scarce and where they go matters; where the budget can relieve most researchers' bottlenecks, the optimal allocation and equal division produce nearly the same research output. The second is how unequally capability is distributed: where a few researchers can produce far more than the rest, getting resources to those few is most of what a funder can add. Every result above is a statement about this quantity: what records and review contribute to capturing it, what seed grants and lotteries give up of it, and how little the timing of spending adds to it.

These results have a practical consequence for funders. No funding mechanism is good or bad in itself. Peer review, seed grants, and lotteries each perform better in some fields than in others, and the two features of the field decide which. A funder choosing among mechanisms should therefore begin by estimating the two features for its own field. The funder can compute the first, its budget relative to researchers' total resources, from its own budget and what it knows about the field. The inequality of capability is harder to observe: the natural proxy is the distribution of research output, but where grants are small, output tracks resources rather than capability, so the output distribution understates capability inequality. A small foundation funding a heavy-tailed field faces the largest value of targeting per dollar. For this foundation, review directed at identifying the few researchers of unusually high capability is worth most, and even noisy review recovers most of that value; seed grants give up the most output; and a lottery among applicants above a review threshold gives up about a sixth of what selection by review gains. A large agency funding an evenly spread field faces a small value of targeting; for this agency, review of any informativeness adds little, seed grants and lotteries give up little, and the best use of the budget is small grants to many researchers.

Three further implications concern peer review. Since most of review's value comes from separating the few researchers of unusually high capability from the rest, and fine distinctions among the many applications of comparable merit add little research output, a review system that spends most of its effort ranking the middle of the distribution spends that effort where the model locates the least output. Second, the finding that percentile scores barely predict productivity comes from funded NIH grants with scores in the top fifth (Fang et al. 2016), and the studies that find modest predictive power also use funded grants (Li and Agha 2015); both

measure prediction among applicants that review has already selected, within the range where the model expects review to add least, so a weak association there is consistent with review of considerable value over the whole applicant pool. Third, a funder that treats review as more reliable than it is loses more output than a funder that treats review as noisier than it is, so a funder uncertain about review’s accuracy should err toward undertrust.

The model holds several things fixed, and its results hold within those limits. There is one funder, and researchers do not respond strategically, so competition among funders, proposal effort, risk choice, and the cost of review are outside the model; the contest models of grant competition cover that ground (Gross and Bergstrom 2019, 2025). Output is a single number per round, so projects do not differ in kind and exploring new directions has no value in the model (Avin 2019b). The review signal is noisy but unbiased; bias and conservatism in review, which are among the strongest arguments for lotteries, are not modeled. Grants are consumed and inputs are complements; the timing results depend on complementarity and the information results do not. The numbers we report are comparisons across the parameter values we sweep, and none is calibrated to a real funder; what applies to a real field is the qualitative features of the dynamics that organize the results.

The model makes three predictions that data on funded research can test. Measures that stand in for capability, such as prior record or review score, should predict research output better the larger a grant is relative to the recipient’s other resources, because on small grants output is nearly independent of capability. How well review scores predict output, measured over the whole applicant pool rather than among funded grants, should be higher in fields whose output distributions have heavier tails. And where funders run lotteries beside review, the output lost to the lottery should be larger in heavier-tailed fields. Each prediction can be tested with data that funders already hold: the first by comparing how well review scores predict output across grant sizes within one applicant pool; the second by comparing that predictive power across fields, scored over the whole applicant pool where the outcomes of funded and unfunded applicants can both be observed; the third by comparing the outputs of lottery-funded and review-funded grants where a funder runs both.

A funder with a fixed budget and imperfect knowledge of researchers faces one problem, and our model gives it one answer. The optimal allocation funds the gap between each researcher’s capability and their resources. Records alone underdetermine that gap; peer review can provide a further signal of capability; and what review is worth, how seed grants and lotteries perform relative to targeted funding, and whether to concentrate or spread all turn on the same two features of a

field. In practically all cases, choosing whom to fund matters more than choosing when. A funder deciding among review, lotteries, and seed grants should therefore begin by asking which kind of field it is funding.

MATERIALS AND METHODS

Model. A funder faces n researchers over T funding rounds. Researcher i has capability K_i and resources R_i ; initial values are drawn from Pareto distributions with tail parameters α_K and α_R (smaller values give more unequal fields) and correlation ρ . In each round, researcher i produces research output drawn from a Poisson distribution with mean $\lambda_i = 2AK_iR_i/(K_i + R_i)$, the harmonic mean of the two inputs scaled by a productivity constant A ; output increases in both inputs and is limited by the scarcer. Between rounds capability compounds, $K_i \leftarrow K_i + \epsilon \lambda_i/A$, at rate ϵ . Resources do not accumulate: $R_i = R_{i0} + g_i$, where the baseline R_{i0} recurs every round and the grant g_i is consumed in the round it is given. The funder’s total budget over the horizon is $bn \mathbb{E}[R_0]$, so at budget scale $b = 1$ it equals the community’s expected baseline resources for one round; the budget does not depend on the horizon.

Information and strategies. The funder knows the distributions but observes no researcher’s capability or resources. It observes each researcher’s output as it accumulates; a resource signal, observed once, gives a noisy estimate of baseline resources (noise τ_R); a review signal, drawn afresh each round it is used, gives a noisy estimate of capability (noise τ_K). Bayesian strategies update beliefs about every researcher after each round and allocate each dollar to the researcher whose expected output, given current beliefs, rises most with it. They differ in information (with or without the review signal) and in planning (myopic, optimizing the current round, or forward-looking, re-planning the remaining budget over the remaining horizon each round). Three baselines complete the set: no funding, against which every result is normalized; uniform funding; and track-record funding, which allocates in proportion to the previous round’s output. Seed grants and lotteries modify these strategies as described in Results (SI Text 2, Table S1).

Parameters and scoring. Tail parameters range from 1.3 to 3.5, covering the Pareto tails near 1 to 2 estimated for research productivity (Lotka 1926; Nicholls 1989) and extending to much more equal fields; the budget scale b from 0.2 to 2, with larger budgets in the timing analyses; the compounding rate ϵ from near zero to 0.85; review noise τ_K from 0.05 to 20; the horizon from one round to ten. Default values are $T = 2$, $n = 50$, $\epsilon = 0.1$, $b = 1$, $\alpha_K = \alpha_R = 2$, $\tau_K = \tau_R = 1$, $\rho = 0$. Each strategy is scored by the sum of expected outputs along its run, which equals realized output averaged over repetitions of the same allocations, and

strategies are compared on identical simulated populations with identical signal and output draws, so comparisons are paired. Results are reported as shares of a named comparison, most often the research output that funding adds: a strategy's expected output minus the expected output under no funding. Analytic results (the gap rule, its corollaries, and the lottery proposition) are proved in SI Text 3 and SI Text 4; the settings behind each figure are in SI Text 7. Code and data are available in the project repository.

SUPPORTING INFORMATION APPENDIX

The SI Appendix holds what the main text states in one sentence or cites without proof: the related literature (SI Text 1), the model in full (SI Text 2), the derivation of the gap rule (SI Text 3), the lottery proposition (SI Text 4), extended results with Figs. S1 to S4 (SI Text 5), the robustness of the results to the production function (SI Text 6), and the simulation specification behind every figure and number (SI Text 7). The table below maps each main-text statement to the section that supports it.

Main-text statement	Where in the SI
The optimal allocation funds the gap: $g_i^* = \max(cK_i - R_{i0}, 0)$	SI Text 3, Proposition 1
Each heuristic can perform worse than uniform division	SI Text 3, Examples 1 and 2
The funded set and every grant increase with the budget; the funding frontier	SI Text 3, Corollary 1
The value of targeting decreases in the budget and vanishes as the budget grows	SI Text 3, Corollary 2; SI Text 5, Fig. S1
Targeting's value relative to uniform funding, across budgets and fields	SI Text 5
The ordering of review's value by field holds from Cobb-Douglas to Leontief	SI Text 6, Fig. S5
Overtrusting review loses more output than undertrusting it	SI Text 5, Fig. S2
Large grants reveal capability; the schedule spends a small share first	SI Text 5, Fig. S3
The shift of spending toward later rounds comes from output produced whether or not funded	SI Text 5
Scheduling's value exists only under complementarity	SI Text 6, Table S2
Equal division outperforms any lottery over the same pool	SI Text 4, Proposition 2
The production function does not settle concentration	SI Text 5, Fig. S4
Model, signals, strategies (Table S1), parameter ranges	SI Text 2
Related literature	SI Text 1
Settings behind every figure and number; code	SI Text 7

SI Text 1. RELATED LITERATURE

Our contribution sits at the intersection of four literatures: the empirical debate over concentrating or dispersing funds, the evidence on how well peer review predicts research outcomes, the case for funding lotteries, and formal models of science funding.

Whether funders should concentrate resources on a few researchers or spread them across many is a long-running dispute, and concentration is the current practice: the top 1% of NIH principal investigators receive close to 10% of research project grant funding, a share that has risen since 1985 (Katz and Matter 2020; Lauer and Roychowdhury 2021). Empirical studies of funded portfolios generally find diminishing marginal returns to funding per researcher, and reviews of this literature conclude that the evidence favors more spreading than current practice (Mongeon et al. 2016; Aagaard et al. 2020). Concentration is also self-reinforcing: early grant winners accumulate advantages beyond the grant itself (Bol et al. 2018). This dispute typically concerns the returns to funding: whether research output per researcher increases or decreases as grants concentrate. In our model, the answer also depends on the size of the funder’s budget relative to researchers’ total resources: with a budget large enough to relieve most bottlenecks, the optimal allocation itself reaches nearly every researcher, so little is lost by spreading funds widely; with a tight budget, effectiveness requires concentrating funds on the researchers with the largest capability-resource gaps (main text). The value of any targeting, in turn, depends on what the funder can know about researchers (main text).

How well peer review predicts research outcomes is contested. Among funded NIH grants, percentile scores predict subsequent productivity barely better than chance (Fang et al. 2016), and reviewers assessing the same application agree only weakly (Pier et al. 2018). Studies with other designs find real but modest predictive power: better scores predict more publications, citations, and patents, including under controls for investigator history and institution (Li and Agha 2015), and a natural experiment in funding finds that higher-scored proposals produce more than lower-scored ones (Park et al. 2015). Both sides of this dispute measure review’s accuracy: how well scores predict outcomes among funded applicants. Our model instead measures review’s value: the expected output that review adds by reducing the funder’s uncertainty about the capability-resource gap. That value depends on the field; the same noisy signal can be worth a great deal in one field and nothing in another (main text), and no single number settles the question.

Doubts about review have made lotteries a live proposal. Advocates argue that randomizing among applications that pass a review threshold saves review costs, reduces bias, and honestly reflects review’s limited ability to discriminate among proposals that pass a review threshold (Fang and Casadevall 2016; Avin 2019a; Roumbanis 2019), and several funders now run partial lotteries (Liu et al. 2020; Heyard et al. 2022; Luebber et al. 2025). The existing formal argument for lotteries measures what a lottery saves: Gross and Bergstrom (2019) model grant

competition as a contest in which scientists invest costly effort in proposals, show that when only a small fraction of proposals is funded this effort can rival the scientific value of the research funded, and show that a partial lottery among proposals above a threshold removes the link between the wasted effort and the fraction funded. What has not been measured is what a lottery gives up: the value of targeting. Our model measures that value. It is negligible where budgets are ample, review is uninformative, or capability is evenly spread, and substantial where capability is heavy-tailed and budgets are tight (main text). Together the two measures let a funder weigh the choice field by field.

Formal models of science funding have concentrated on the incentive costs of competition and on the value of exploration. Beyond the contest model above, Gross and Bergstrom (2025) find, in a game-theoretic model of contests for scarce rewards, that intensifying competition pushes scientists toward higher-risk, higher-return projects. Avin (2019b) simulates funding strategies on an epistemic landscape and finds that allocation with an explicitly random component significantly outperforms a peer-review-like strategy on large landscapes, because review concentrates effort on regions already known to be productive. Sorzano and Pueche-Granados (2026) model allocation as a decision under heavy-tailed uncertainty about project value, and combine a citation-based signal with a lottery weighted by that signal. In each of these models, funding raises a single quantity, a project’s chance of success or a researcher’s productivity, and none treats the timing of spending as the funder’s choice. In our model, a researcher’s research output increases in both capability and resources but is limited by the scarcer of the two, and the funder observes neither directly. This formulation lets us address questions raised by all four literatures: how widely funds should be spread, what peer review is worth, and when lotteries lose little.

SI Text 2. THE MODEL IN FULL

A funder faces a population of n researchers over T funding rounds. Researcher i has capability K_i and resources R_i . Initial capabilities and resources are drawn from Pareto distributions with tail parameters α_K and α_R ; smaller tail parameters produce more unequal fields, and a correlation parameter ρ lets capability and resources be correlated across researchers. The funder knows these distributions; it never observes any researcher’s capability or resources directly.

In each round, researcher i produces research output¹ at random, in discrete units, with expected output²

$$\lambda_i = A \Lambda(K_i, R_i),$$

where A is a productivity constant and Λ is a production function that increases in both arguments but is limited by the scarcer of the two. Our main analysis takes Λ to be the harmonic mean:

$$\lambda_i = \frac{2AK_iR_i}{K_i + R_i}.$$

The harmonic mean makes capability and resources complements without making either an absolute requirement: a capable researcher with few resources still produces, by working more slowly or borrowing equipment, but no surplus of one input fully substitutes for a shortage of the other. SI Text 6 repeats our analysis for a family of production functions running from Cobb-Douglas, under which either input can substitute for the other, to Leontief, under which neither can.

Capabilities increase through research. Between rounds,

$$K_i \leftarrow K_i + \epsilon \Lambda(K_i, R_i),$$

where ϵ is the compounding rate: doing research builds expertise, and the same bottleneck that limits output limits growth. Growth requires both inputs, and capability never declines. Resources, by contrast, do not accumulate: resources in a round are baseline plus grant, $R_i = R_{i0} + g_i$. The baseline R_{i0} recurs every round, representing the time and support a researcher has independent of the funder, while grants are consumed in the round they are given. Whatever lasts from a grant (skills, publications, standing) enters the model as capability.

The funder's total budget is fixed before any outcomes are observed. We parameterize it by a budget scale b : at $b = 1$, the budget equals the community's expected baseline resources for one round, and the budget scales linearly in b (SI Text 7). The budget does not grow with the horizon: a longer horizon varies the timing of spending, never its amount. Strategies that do not plan ahead spend an equal share of the budget each round. Forward-looking strategies may spend on any schedule: each round, they re-plan the entire remaining budget over the remaining horizon and execute the current round's allocation.

The funder learns as it goes. It observes each researcher's track record, the output they have produced to date, as it accumulates, and two further signals are available. A resource signal, observed once at the start, gives a noisy estimate of each

¹ We are agnostic about the interpretation of output units: they may be interpreted as publications, as findings, or as units of scientific value.

² Output in each round is drawn from a Poisson distribution with rate λ_i , independently across researchers and rounds given the state.

Strategy	Information	Planning
No funding	none	none
Uniform	none	none
Track record	track record	none
Myopic, without review	track record + resource signal	current round
Forward, without review	track record + resource signal	remaining horizon
Myopic, with review	track record + resource signal + review	current round
Forward, with review	track record + resource signal + review	remaining horizon

TABLE S1. The funding strategies compared.

researcher’s baseline resources (noise τ_R), as one might obtain from institutional information. A review signal, drawn afresh in each round that a strategy uses it, gives a noisy estimate of capability (noise τ_K), as one might obtain from proposal evaluation and expert judgment. The review signal’s informativeness is governed by τ_K : the smaller the noise, the more informative the review. Bayesian strategies update beliefs about every researcher after each round and allocate each dollar to the researcher whose expected output, given the funder’s current beliefs, rises most with it.

We compare the strategies in Table S1. The Bayesian strategies differ along two dimensions: information, with or without the review signal, and planning, myopic (optimizing the current round’s expected output) or forward-looking (optimizing expected output over the whole remaining horizon). Three baselines complete the set: no funding, against which every result is normalized; uniform funding, which splits each round’s budget equally; and track-record funding, which allocates in proportion to the previous round’s output. Seed grants and lotteries, which modify these strategies, are introduced in the main text.

Each of our conclusions rests on a proof or on simulations across the following parameter ranges; default values and the settings behind each figure are in SI Text 7. The tail parameters α_K and α_R range from 1.3 to 3.5. Empirical distributions of research productivity have Pareto tail parameters near 1 to 2 (Lotka 1926; Nicholls 1989); our range covers these values and extends to much more equal fields. The budget scale b ranges from 0.2 to 2, from a funder whose budget is small beside the community’s own resources to one able to relieve most researchers’ bottlenecks; the main text and the main text also examine larger budgets. The compounding rate ϵ ranges from near zero, where funding raises only current output, to 0.85, where this round’s funding substantially raises later capability. The review noise τ_K ranges from 0.05, nearly perfect review, to 20, nearly uninformative review, a range wide enough to contain the noise implied by empirical estimates of review’s predictive accuracy (main text). The correlation ρ ranges from -0.5 to 0.8 , and the horizon T from one round to ten.

Randomness enters through the initial population draws, the signals, and realized output; capability growth follows expected output, so a researcher's accumulation depends on their capability and resources rather than on luck in a given round. We score each strategy by expected output and compare strategies on identical simulated populations, and we report each strategy's improvement over no funding in percent (SI Text 7).

With the model in place, the funder's problem is well defined: choose grants, given beliefs, to maximize expected total output over the horizon. Next we solve the case in which the funder has complete information regarding researchers' capabilities and resources.

SI Text 3. DERIVATION OF THE GAP RULE

The within-round allocation problem. A population of n researchers. Researcher i has capability K_i and baseline resources R_{i0} , both fixed within the round. A grant $g_i \geq 0$ adds to researcher i 's resources for the round, so that their expected output is $A \Lambda(K_i, R_{i0} + g_i)$ with $\Lambda(K, R) = 2KR/(K + R)$, the production function of § SI Text 2, and $A > 0$ the common productivity scale. Write

$$\varphi_i(g) = \frac{2A K_i (R_{i0} + g)}{K_i + R_{i0} + g}$$

for researcher i 's expected output as a function of their grant alone.

Throughout, $A > 0$, $K_i > 0$ and $R_{i0} \geq 0$ for every i , and the round's budget is $B > 0$. Nothing else about the model is used, so the result holds for every realized population and every round's budget; across rounds, the constant c and the funded set are recomputed from the new population state.

The funder observes $(K_i, R_{i0})_{i=1}^n$ and divides the budget so as to maximize expected output:

$$(P) \quad \max_{g_1, \dots, g_n \geq 0} \sum_{i=1}^n \varphi_i(g_i) \quad \text{subject to} \quad \sum_{i=1}^n g_i \leq B.$$

Three lemmas.

Lemma 1 (the return to a grant). *For each i , on $g \in [0, \infty)$: (i) φ_i is strictly increasing, with $\varphi'_i(g) = 2A K_i^2 / (K_i + R_{i0} + g)^2 > 0$; (ii) φ_i is strictly concave: $\varphi''_i(g) = -4A K_i^2 / (K_i + R_{i0} + g)^3 < 0$; (iii) $\varphi'_i(g) < 2A$ whenever $R_{i0} + g > 0$, and $\varphi_i(g) \uparrow 2AK_i$ as $g \rightarrow \infty$.*

Proof. Write $S = K_i + R_{i0} + g$. Then $\varphi_i(g) = 2AK_i(S - K_i)/S = 2AK_i - 2AK_i^2/S$, and (i) and (ii) follow by differentiating in g (note $dS/dg = 1$). For (iii):

$\varphi'_i(g) = 2AK_i^2/S^2 < 2A$ exactly when $S > K_i$, that is, when $R_{i0} + g > 0$; and $\varphi_i(g) = 2AK_i - 2AK_i^2/S \rightarrow 2AK_i$ as $S \rightarrow \infty$. $\square$

Lemma 1 says that the marginal value of a dollar to researcher i increases with their capability and decreases with the resources they already hold, that each dollar granted lowers the value of the next, and that expected output never exceeds $2AK_i$.

Lemma 2 (existence, uniqueness, exhaustion). *Problem (P) has exactly one solution $g^* = (g_1^*, \dots, g_n^*)$, and it exhausts the budget: $\sum_i g_i^* = B$.*

Proof. The feasible set is compact and convex and the objective $f(g) = \sum_i \varphi_i(g_i)$ is continuous and, by lemma 1(ii), strictly concave, so a maximizer exists and is unique. If $\sum_i g_i^* < B$, increasing any one g_i^* slightly stays feasible and raises f (lemma 1(i)), contradicting optimality. $\square$

Lemma 3 (equal marginal value). *A feasible allocation g with $\sum_i g_i = B$ solves (P) if and only if there is a $\nu > 0$ such that*

$$\varphi'_i(g_i) = \nu \text{ for every } i \text{ with } g_i > 0, \quad \varphi'_i(0) \leq \nu \text{ for every } i \text{ with } g_i = 0.$$

Proof. Necessity. Let g^* solve (P). Since $B > 0$ is exhausted (lemma 2), some researcher is funded: $g_i^* > 0$ for some i . Fix any such i and any $j \neq i$, and for $0 < \delta \leq g_i^*$ consider the transfer that replaces g_i^* by $g_i^* - \delta$ and g_j^* by $g_j^* + \delta$, leaving the rest unchanged. The result is feasible, so optimality gives

$$\varphi_j(g_j^* + \delta) - \varphi_j(g_j^*) \leq \varphi_i(g_i^*) - \varphi_i(g_i^* - \delta).$$

Dividing by δ and letting $\delta \downarrow 0$ yields $\varphi'_j(g_j^*) \leq \varphi'_i(g_i^*)$. If j is also funded, the same argument with i and j exchanged gives the reverse inequality, so all funded researchers share a common marginal value; call it $\nu = \varphi'_i(g_i^*) > 0$ (positivity by lemma 1(i)). For unfunded j , the displayed inequality reads $\varphi'_j(0) \leq \nu$.

Sufficiency. Let g satisfy the condition with multiplier ν , and let h be any feasible allocation. Concavity of φ_i gives, for every i , $\varphi_i(h_i) \leq \varphi_i(g_i) + \varphi'_i(g_i)(h_i - g_i)$. For funded i , $\varphi'_i(g_i) = \nu$, so $\varphi'_i(g_i)(h_i - g_i) = \nu(h_i - g_i)$. For unfunded i , $g_i = 0$ and $h_i - g_i = h_i \geq 0$, so $\varphi'_i(0)(h_i - g_i) \leq \nu(h_i - g_i)$. Summing over i :

$$f(h) \leq f(g) + \nu \sum_i (h_i - g_i) = f(g) + \nu \left(\sum_i h_i - B \right) \leq f(g),$$

since h is feasible and $\nu > 0$. So g solves (P). $\square$

Lemma 3 is the formal content of the sentence in the main text: an allocation is optimal exactly when no dollar can be moved to a researcher who can produce more value with it.

The proposition.

Proposition 1 (the gap rule). *The unique solution of (P) is*

$$g_i^* = \max(c K_i - R_{i0}, 0), \quad i = 1, \dots, n,$$

where $c > 0$ is the unique constant satisfying the budget identity $\sum_{i=1}^n \max(c K_i - R_{i0}, 0) = B$. Equivalently, $c = \sqrt{2A/\nu} - 1$, where $\nu \in (0, 2A)$ is the common marginal value of lemma 3.

Proof. Let g^* be the unique solution (lemma 2) and ν the multiplier that lemma 3 associates with it. The proof proceeds in four steps.

Step 1: $\nu \in (0, 2A)$, so $c = \sqrt{2A/\nu} - 1$ is well defined and positive. Positivity of ν is part of lemma 3. For the upper bound: some researcher i is funded, and for them $R_{i0} + g_i^* \geq g_i^* > 0$, so lemma 1(iii) gives $\nu = \varphi'_i(g_i^*) < 2A$. Then $2A/\nu > 1$, so $c = \sqrt{2A/\nu} - 1 > 0$.

Step 2: funded researchers receive exactly their gap. Let $g_i^* > 0$. By lemma 3, $2A K_i^2 / (K_i + R_{i0} + g_i^*)^2 = \nu$. Solving, and taking the positive square root (both sides of the next display are positive),

$$K_i + R_{i0} + g_i^* = K_i \sqrt{2A/\nu} = (1 + c) K_i,$$

so $g_i^* = c K_i - R_{i0}$. Since $g_i^* > 0$, this equals $\max(c K_i - R_{i0}, 0)$.

Step 3: unfunded researchers have no gap. Let $g_i^* = 0$. By lemma 3, $2A K_i^2 / (K_i + R_{i0})^2 = \varphi'_i(0) \leq \nu$, so $K_i + R_{i0} \geq K_i \sqrt{2A/\nu} = (1 + c) K_i$, that is, $R_{i0} \geq c K_i$. Then $\max(c K_i - R_{i0}, 0) = 0 = g_i^*$.

Steps 2 and 3 establish $g_i^* = \max(c K_i - R_{i0}, 0)$ for every i , and the budget identity is exhaustion (lemma 2). It remains to show c is the only constant satisfying that identity.

Step 4: the budget identity pins c uniquely. Define $G(c) = \sum_{i=1}^n \max(c K_i - R_{i0}, 0)$ for $c \geq 0$. G is continuous and nondecreasing, with $G(0) = 0$ and $G(c) \rightarrow \infty$ as $c \rightarrow \infty$. Moreover G is strictly increasing wherever it is positive: if $G(c) > 0$, then $c K_j > R_{j0}$ for some j , and for any $c' > c$, $G(c') \geq G(c) + K_j (c' - c) > G(c)$. Now suppose $c_1 < c_2$ both satisfy $G(c) = B$. Since $B > 0$, $G(c_1) > 0$, so $G(c_2) > G(c_1) = B$, a contradiction. $\square$

Consequences.

Corollary 1 (funding frontier and budget comparative statics). *(i) Researcher i is funded, $g_i^* > 0$, exactly when $K_i > R_{i0}/c$. (ii) As a function of the budget B , the constant $c(B)$ is continuous and strictly increasing; hence the funded set is nondecreasing in B , every funded researcher's grant $c(B)K_i - R_{i0}$ is strictly increasing in B , and the common marginal value $\nu(B) = 2A/(1+c(B))^2$ is strictly decreasing in B . Both parts follow from the formula and Step 4.*

Corollary 2 (the value of targeting vanishes in the budget). *Let $f^*(B)$ denote the optimal value of (P) and $f^u(B) = \sum_i \varphi_i(B/n)$ the value of uniform funding. Then $0 \leq f^*(B) - f^u(B)$ for every B , and $f^*(B) - f^u(B) \rightarrow 0$ as $B \rightarrow \infty$.*

Proof. The first inequality holds because uniform funding is feasible. For the second: by lemma 1(iii), $\varphi_i(g) < 2AK_i$ for every g , so $f^*(B) \leq 2A \sum_i K_i$; and $f^u(B) = \sum_i \varphi_i(B/n) \rightarrow 2A \sum_i K_i$ as $B \rightarrow \infty$. Hence $0 \leq f^*(B) - f^u(B) \leq 2A \sum_i K_i - f^u(B) \rightarrow 0$. $\square$

Since uniform funding's own gain over no funding does not vanish (it approaches $2A \sum_i K_i - \sum_i \varphi_i(0) > 0$), the same conclusion holds for the value of targeting measured relative to that gain, the quantity plotted in Fig. S1.

Example 1 (funding the track record can produce less than uniform funding). *Two researchers, $A = 1$, budget $B = 2$. Researcher 1: $K_1 = 8$, $R_{10} = 8$ (productive and well resourced; expected output $\varphi_1(0) = 8$). Researcher 2: $K_2 = 4$, $R_{20} = 1$ (capable but with few resources; $\varphi_2(0) = 8/5$). Allocating in proportion to expected output gives $g = (5/3, 1/3)$ and total output $570/53 \approx 10.75$. Uniform funding, $g = (1, 1)$, gives $568/51 \approx 11.14$. The gap rule ($c = 3/4$) gives $g^* = (0, 2)$ and $80/7 \approx 11.43$.*

Example 2 (funding the under-resourced can produce less than uniform funding). *Two researchers, $A = 1$, budget $B = 2$. Researcher 1: $K_1 = 10$, $R_{10} = 5$. Researcher 2: $K_2 = 1/2$, $R_{20} = 0$ (the scarcest resources in the population). Directing the budget to the researcher with the fewest resources gives total output $112/15 \approx 7.47$. Uniform funding gives $49/6 \approx 8.17$. The gap rule ($c = 2/3$) gives $g^* = (5/3, 1/3)$ and $42/5 = 8.4$.*

SI Text 4. LOTTERIES AND EQUAL DIVISION

Fix a round, a budget $B > 0$, and a pool $P \subseteq \{1, \dots, n\}$ of m researchers. The *equal division* over P grants each researcher in P the amount B/m . A *lottery* over P with $k \leq m$ winners selects k researchers from P uniformly at random and grants each winner B/k . Expected output is evaluated as in SI Text 3: given an

allocation g , it is $\sum_i \varphi_i(g_i)$, and for a random allocation, the expectation is taken over the draw as well.

Proposition 2 (lotteries are outperformed by equal division). *For every pool P , budget B , and winner count $k \leq m$, the lottery’s expected output does not exceed the equal division’s, with equality only if $k = m$. In particular, a lottery over the whole population, at any winner count, produces at most uniform funding’s expected output.*

Proof. Researchers outside P receive nothing under either scheme. For $i \in P$, the lottery’s grant to i is a random variable $\tilde{g}_i$ equal to B/k with probability k/m and 0 otherwise, so $\mathbb{E}[\tilde{g}_i] = B/m$. By linearity, the lottery’s expected output is $\sum_{i \notin P} \varphi_i(0) + \sum_{i \in P} \mathbb{E}[\varphi_i(\tilde{g}_i)]$. Since φ_i is strictly concave (Lemma 1(ii)), Jensen’s inequality gives $\mathbb{E}[\varphi_i(\tilde{g}_i)] \leq \varphi_i(\mathbb{E}[\tilde{g}_i]) = \varphi_i(B/m)$ for every $i \in P$, with equality only if $\tilde{g}_i$ is degenerate, that is, $k = m$. Summing over i yields the claim; taking $P = \{1, \dots, n\}$ yields the comparison with uniform funding. $\square$

Remark 1 (what the proposition does and does not cover). *The proposition compares expected output within a fixed pool. It measures the effect of the randomization alone: whatever pool review selects, replacing the lottery within that pool by the equal division of the same money weakly increases expected output. The proposition says nothing about the choice of pool, about the review ranking within the pool that the equal division also ignores, or about anything a lottery saves outside the model (review costs, proposal effort, bias). The main text measures the first two and cites the literature on the third.*

SI Text 5. EXTENDED RESULTS

This section gives the full treatment of results that the main text states in one sentence. Each subsection opens with the main-text statement it supports.

The value of targeting decreases in the budget. *Main text:* “The value of targeting, the expected output the optimal allocation adds over equal division, decreases in the budget and vanishes as the budget grows.”

How much targeting matters depends on the budget. A researcher’s expected output increases with every dollar of resources, but with diminishing marginal returns, and it never exceeds $2AK_i$: expected output is bounded above by a quantity set by capability, which it approaches but does not attain. When the budget is large, optimal and uniform funding therefore differ little in the output they produce: under either, every researcher’s output is near its upper bound, and the difference between the two vanishes as the budget increases (Corollary 2, SI Text

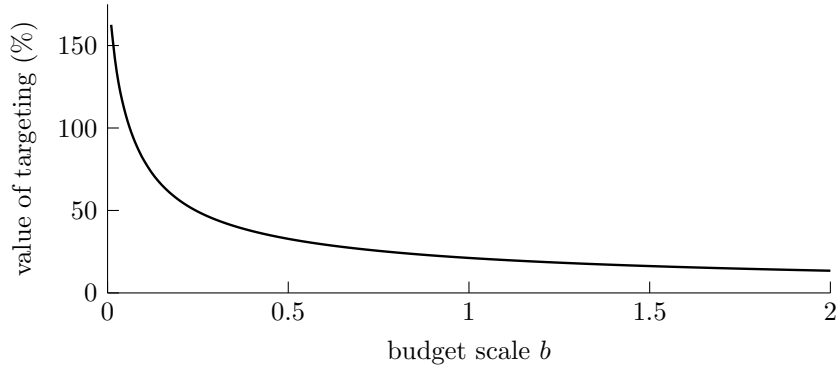


FIGURE S1. The value of targeting decreases in the budget. The additional expected output of the optimal allocation over uniform funding, as a fraction of uniform funding’s gain over no funding, against the budget scale b , for one population drawn from the default distributions (Corollary 2).

3). When the budget is small, dollars are scarce, researchers differ in the marginal value of a dollar, and output depends on where each dollar goes. Targeting matters where the budget is small relative to researchers’ resources and little where the budget is large (Fig. S1). This dependence on the budget recurs later: it sets how much output is lost when seed grants and lotteries override the optimal allocation, and it determines when optimal funding concentrates rather than spreads (both in the main text).

The value of targeting relative to uniform funding. *Main text:* “For a funder allocating with records and review, targeting adds more research output than uniform funding’s entire gain over no funding in a heavy-tailed field with reliable review and a tight budget, and about a quarter of that gain in the default field at the largest budget we sweep.”

The review-informed funder’s gain over uniform funding, as a fraction of uniform funding’s gain over no funding, decreases from 1.17 to 0.72 across budget scales $b = 0.2$ to 2 with heavy-tailed capability ($\alpha_K = 1.3$) and reliable review ($\tau_K = 0.3$), and from 0.74 to 0.25 in the default field with the default signal ($\alpha_K = 2$, $\tau_K = 1$). Two rounds; 200 simulated populations per point (the populations of Fig. 4A). Giving up targeting loses little output under the same conditions that make review worth little: where capability is evenly spread, the budget is large, or review is uninformative.

Overtrusting review. *Main text:* “A funder that overtrusts review, treating a noisy signal as reliable, can do worse than one that ignores review altogether, and overtrusting review loses more output than undertrusting it.”

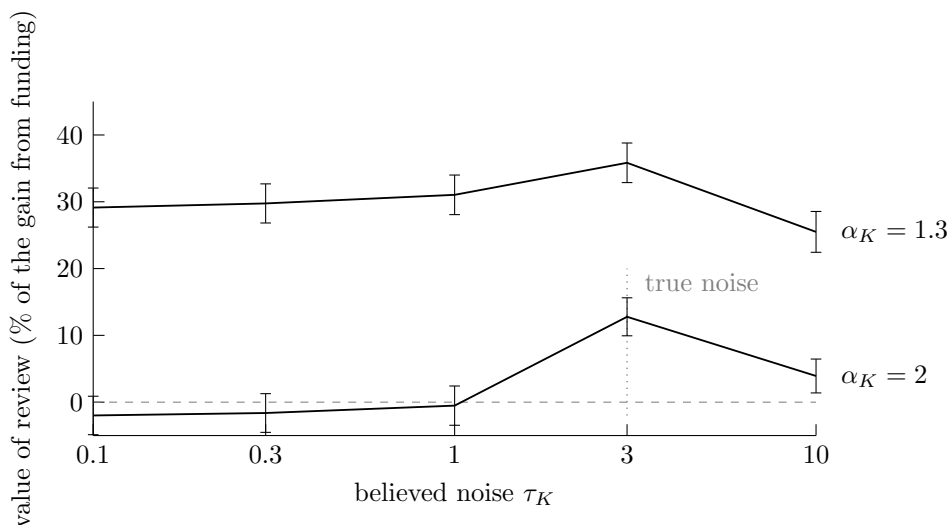


FIGURE S2. Overtrusting review loses more research output than undertrusting it. The value of review as a percent of the research output that a correctly calibrated review-informed funder adds over no funding, as the funder’s believed signal noise varies while the true noise is $\tau_K = 3$ (dotted rule); bars are one standard error. At the default capability distribution, overtrust takes the value below zero. Settings: SI Text 7.

One condition qualifies these results: the funder must weight review at its true informativeness. A funder that overtrusts review, treating a noisy signal as reliable, can do worse than one that ignores review altogether: with moderately spread capability, overtrust turns review’s value negative; with heavy-tailed capability it reduces the value without reversing it (Fig. S2).³ Overtrusting review loses more output than undertrusting it.

Grant size, learning, and the schedule. *Main text:* “The shift of spending toward later rounds comes from the output researchers produce whether or not they are funded, which compounds and reveals capability while the funder waits.”

How much a funder can learn from its own grants depends on their size. A researcher on a small grant produces output nearly proportional to the grant and nearly independent of capability, so a funder whose grants are small learns almost nothing it can exploit later: in a field whose researchers have few resources, with no review signal, such a funder gains almost nothing over uniform funding. Large grants change this. As grants approach the scale of capability itself, output separates the capable from the rest, and the funder recovers most of the value of

³ A funder that overtrusts tenfold (true $\tau_K = 3$, belief 0.3) loses more than review’s full value at the default capability distribution; a funder that undertrusts tenfold (true $\tau_K = 0.3$, belief 3) retains over half of it.

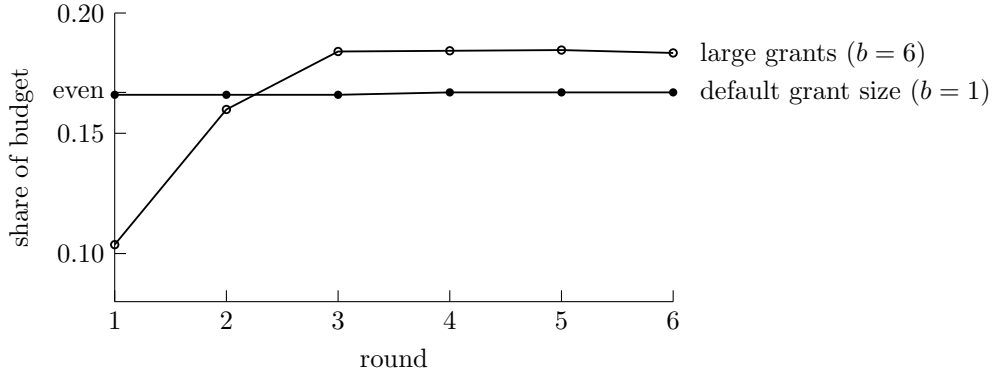


FIGURE S3. With large grants the funder spends a small share first and the rest once it has observed research output. The share of the budget spent in each of six rounds, in a field whose researchers have few resources, with heavy-tailed capability and no review signal, at the default grant size (filled) and at six times that size (open). Settings: SI Text 7.

discrimination, on a schedule that spends a small share early, so that output can be observed, and the rest once the funder has learned who is capable (Fig. S3).⁴

The shift of spending toward later rounds comes from the output researchers produce whether or not they are funded: that output compounds and reveals capability while the funder waits. Growth from grant-added output alone would shift the schedule slightly toward earlier rounds, since earlier grants have longer to compound.⁵

Concentration and the production function. *Main text:* “Stronger complementarity between capability and resources concentrates the review-informed funder’s grants when the budget is tight in a heavy-tailed field, spreads them when the budget is ample, and has a non-monotone effect where capability is evenly spread.”

Seed grants and lotteries both spread funds more widely than targeting does, and both raise the older question of whether funding should be concentrated among a few researchers or spread across many. One might hope that the production function settles this question: the harder it is to substitute resources for capability, the more the money should go to the few researchers who can use it. To test this,

⁴ In a community whose researchers have few resources, with heavy-tailed capability and no review signal, over six rounds, the Bayesian funder’s gain over uniform funding increases from a third of one percent of uniform funding’s output at the default grant size to nine percent at six times that size and ten percent at twelve times, roughly the gain a reliable review signal delivers at the default size.

⁵ With growth from unfunded output switched off, the schedule’s center of mass is 0.48 to 0.49, below even; with growth from grant-added output switched off, 0.53 to 0.68 (five rounds).

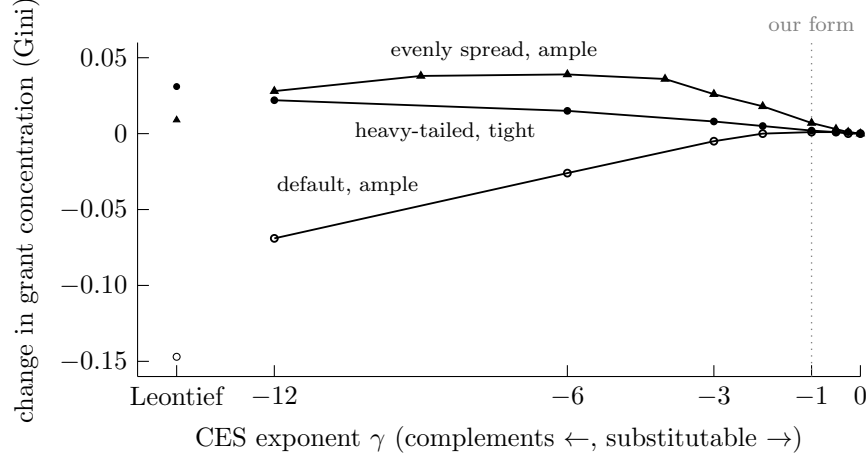


FIGURE S4. Complementarity concentrates funding only when the budget is small. The change in the Gini coefficient of the review-informed funder’s grants, relative to its Cobb-Douglas level, as the production function moves from Cobb-Douglas ($\gamma = 0$) toward Leontief (detached marks at the left edge), for three field-and-budget combinations. Settings: SI Text 7.

we embed our production function in a family indexed by an exponent γ , from Cobb-Douglas ($\gamma = 0$), under which a shortfall in either input can be offset by more of the other, through our form ($\gamma = -1$), to Leontief, under which output depends only on the scarcer input; the harder it is to substitute one input for the other, the stronger the complementarity (SI Text 6). The production function does not settle the question (Fig. S4). With a tight budget in a heavy-tailed field, stronger complementarity does concentrate the review-informed funder’s grants. With an ample budget, stronger complementarity spreads them, a pattern consistent with a funder that, once the budget allows, brings every researcher’s resources up toward the level their capability can use. Where capability is evenly spread and the budget ample, the relation is not even monotone: concentration peaks at an intermediate complementarity and declines toward both ends.⁶ How widely funds should be spread is set jointly by the production function, the budget, and the field’s capability distribution; no one of the three decides it alone.

⁶ Gini coefficients of the review-informed funder’s grants, from Cobb-Douglas to Leontief: 0.92 to 0.95 in a heavy-tailed field with a tight budget; 0.35 to 0.21 in the default field with an ample budget; 0.63 to 0.45 in a heavy-tailed field with an ample budget. In the evenly spread field with an ample budget the Gini coefficient is 0.16 at Cobb-Douglas, peaks at 0.196 near $\gamma = -6$, and falls to 0.185 at $\gamma = -12$ (3.5 standard errors above the $\gamma = -12$ level).

SI Text 6. ROBUSTNESS TO THE PRODUCTION FUNCTION

This appendix asks which of our results depend on the harmonic-mean production function of SI Text 2. We embed that function in a family indexed by an exponent γ : expected output is $A M_\gamma(K, R)$, where $M_\gamma(K, R) = ((K^\gamma + R^\gamma)/2)^{1/\gamma}$ is the power mean of capability and resources. At $\gamma = -1$ the power mean is the harmonic mean $2KR/(K + R)$, our form. In the limit $\gamma \rightarrow 0$ (Cobb-Douglas) it is the geometric mean $\sqrt{KR}$, and a shortfall in either input can be offset by more of the other; as γ decreases the inputs grow harder to substitute; in the limit $\gamma \rightarrow -\infty$ (Leontief) it is $\min(K, R)$, and output depends only on the scarcer input. We re-run the model's central comparisons across this family.⁷

The value of the review signal survives the whole family (Fig. S5). At every production function tested, from Cobb-Douglas to Leontief, the signal's value increases as capability grows more heavy-tailed and as review grows more informative. The ordering of review's value by field (main text) is therefore not an artifact of our choice of production function. What changes with the production function is the scale. The more complementary the inputs, the more review is worth: under heavy-tailed capability with reliable review, the signal's value increases from about 16 percent of the research output that funding with review adds at Cobb-Douglas to about 47 percent at $\gamma = -3$ and about half at Leontief.⁸ Complementarity raises what targeting is worth, and with it the value of the information that targeting requires.

The timing results of the main text do not survive the substitutable end of the family (Table S2). Under Cobb-Douglas production the value of scheduling is indistinguishable from zero at every compounding rate, and the optimal schedule stays within about a hundredth of even. At $\gamma = -3$, a production function more complementary than ours, both the value of scheduling and the shift of spending toward later rounds increase with the compounding rate and exceed their Cobb-Douglas values at every rate. The timing results hold where capability and resources are complements and vanish where they are substitutable.

How widely the informed funder spreads its grants across this family is reported in the main text (Fig. S4).

⁷ Settings: SI Text 7.

⁸ 15.8, 47.1, and 51.4 percent. Cobb-Douglas lies below the two more complementary production functions at every tested tail and noise level at which the value is distinguishable from zero; $\gamma = -3$ and Leontief are within one standard error of each other at the noisiest review under heavy tails.

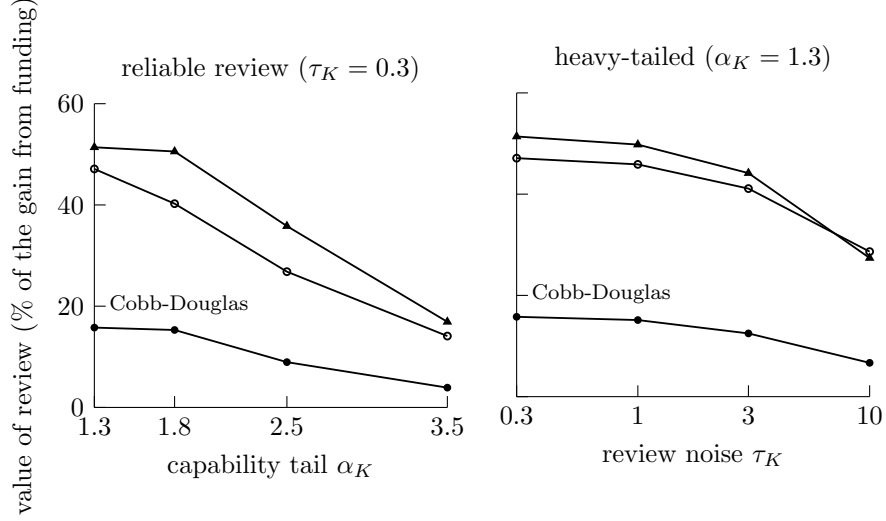


FIGURE S5. The structure of review’s value survives the production function. The value of the review signal, as a percent of the research output that funding with review adds over no funding, across the family of production functions: at every production function tested, the value increases as capability grows more heavy-tailed (left) and as review grows more informative (right), and its scale increases with complementarity. Filled circles: Cobb-Douglas; open circles: $\gamma = -3$; triangles: Leontief. Settings: SI Text 7.

SI Text 7. SIMULATION SPECIFICATION

Budget and productivity. Over the whole horizon, the funder’s total budget is $bn \mathbb{E}[R_0]$, where $\mathbb{E}[R_0]$ is the expected baseline resources of one researcher, so at $b = 1$ the budget equals the community’s expected baseline resources for one round. The budget does not depend on the horizon T . Strategies that do not plan ahead spend $1/T$ of the budget each round. For the timing runs of the main text (Figs. S3 and Fig. 3A) the budget is scaled against capability instead, $bn \mathbb{E}[K_0]$, so that a community with few resources does not zero the budget. The simulations set the productivity constant so that expected output is $K_i R_i / (K_i + R_i)$, that is, $A = 1/2$ in the notation of SI Text 2. The code parameterizes the budget as $2b' n \mathbb{E}[R_0]$; its b' is half of the b reported here.

Default parameters. Unless a figure’s settings below say otherwise: $T = 2$ rounds; $n = 50$ researchers; capability and baseline resources drawn from Pareto distributions with scale 1 and tail parameters $\alpha_K = \alpha_R = 2$; correlation $\rho = 0$; compounding rate $\epsilon = 0.1$; budget scale $b = 1$; review noise $\tau_K = 1$ and resource-signal noise $\tau_R = 1$; seed-grant fraction 0.25 where seed grants are used.

TABLE S2. The value of scheduling and the optimal schedule by production function. Five rounds. The value of scheduling is the forward-looking funder’s gain over the myopic funder, both with review, as a percent of the research output that the forward-looking funder’s funding adds; the schedule’s center of mass is 0.5 when spending is even and larger when spending shifts toward later rounds. One hundred simulated populations per point; standard errors of the gain are at most 0.01 (Cobb-Douglas) and 0.4 ($\gamma = -3$) percentage points.

compounding rate ϵ	0.05	0.15	0.30	0.55	0.85
<i>Value of scheduling (percent of the gain from funding)</i>					
Cobb-Douglas ($\gamma = 0$)	-0.01	-0.01	0.00	0.01	0.03
$\gamma = -3$	0.05	0.56	1.96	4.56	6.52
<i>Center of mass of the optimal schedule</i>					
Cobb-Douglas ($\gamma = 0$)	0.499	0.500	0.501	0.505	0.512
$\gamma = -3$	0.511	0.540	0.577	0.623	0.660

Scoring and comparison. Each strategy is scored by the sum of expected outputs along its run, which equals realized output averaged over repetitions of the same allocations; this removes simulation noise from comparisons without favoring any strategy. Every configuration is run on independently drawn populations, with the same random draws (populations, signals, and realized output) used for every strategy, so that comparisons between strategies are paired. Results are reported as shares of a named comparison, stated in each figure: most often the research output that funding adds (a strategy’s expected output minus the expected output under no funding), and otherwise the records-only shortfall or the optimal allocation’s gain. Standard errors, where shown, are across simulated populations.

Settings by figure.

Code. The model, the sweeps, and the scripts that produce every figure and number in the paper are in the project repository; the archived version will be cited at publication.

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Figure	Settings that differ from the defaults
Fig. 2 extitC	Review noise τ_K from 0.05 to 20; $\alpha_K \in \{1.3, 2, 3.5\}$; 50 populations per point. The records-only shortfall relative to complete information is 40, 27, and 12 percent of the research output that complete-information funding adds, at the three tail parameters.
Fig. 2 extitB	$T = 20$; 50 populations. Records-only is the myopic funder without review; with review is the myopic funder with review. Under heavy-tailed capability ($\alpha_K = 1.3$) the correlations run 0.07 to 0.43 against 0.92 in round one.
Fig. S2	True noise $\tau_K = 3$; believed noise from 0.1 to 10; $\alpha_K \in \{1.3, 2\}$; 200 populations. Each negative value is within two standard errors of zero.
Fig. S3	$T = 6$; community with few resources (Pareto scale for resources 0.001); $\alpha_K = 1.3$; no review signal ($\tau_K = 100$); $\epsilon = 10^{-4}$; budget scaled against capability; $b = 1$ (50 populations) and $b = 6$ (24 populations, standard error of each share at most 0.006).
Fig. 3 extitA	$T = 5$; forward-looking funder with review; budget scaled against capability; resource scales 0.001, 0.3, and 3; ϵ from 10^{-4} to 0.85; 200 populations. Five resource scales were run; the figure shows three, and the other two lie between them.
Fig. 3 extitB	T from 2 to 10 (to 5 for $\epsilon \in \{0.05, 0.15, 0.55\}$); $\epsilon \in \{0.05, 0.15, 0.3, 0.55, 0.85\}$; 64 to 200 populations per point. The gain is the forward-looking funder's output over the myopic funder's, both with review; negative gains, all within noise of zero, are shown as zero.
Fig. 4 extitA	Review-informed forward-looking funder; seed-grant fraction from 0 to 0.75; heavy-tailed field ($\alpha_K = 1.3$) with reliable review ($\tau_K = 0.3$) at $b \in \{0.2, 1, 2\}$, and the default field at $b = 1$; 200 populations. The output lost is reported as a share of the review-informed funder's gain over no funding in the same setting.
Fig. 4 extitB	Single round; $n = 50$; a fifth of the population funded; review score K_i plus Gaussian noise of standard deviation τ ; expected output computed exactly given each allocation; 2,000 populations per point (Monte Carlo standard error at most 0.002). Left: $\alpha_K = 1.3$, $b = 0.2$. Right: $\alpha_K = 3.5$, $b = 1$.
Fig. S5, Table S2	Production functions with $\gamma \in \{0, -3\}$ and Leontief; 100 populations per point (50 for Leontief); $T = 2$ for the signal value and $T = 5$ for the timing table. These runs use an earlier version of the allocation routine, so their magnitudes are compared with one another and not with the body's figures. Leontief allocations use a finer allocation step, with ties at the kink broken by fill order.
Fig. S4	Single round; review-informed funder; γ from 0 to -12 and Leontief; three field-and-budget combinations ($\alpha_K = 1.3$, $b = 0.2$; $\alpha_K = 2$, $b = 2$; $\alpha_K = 3.5$, $b = 2$); 200 populations per point, 400 at the refinement points of the evenly spread curve.

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